Physics of the Earth and Planetary Interiors, 52 (1988) 243–255
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Palaeozoic reconfigurations of north Scotland based on palaeomagnetic results from Orkney dykes

K.M. Storetvedt

Institute of Geophysics, University of Bergen, 5000 Bergen (Norway)

L.E. Otterå

Bergen Maskinisthøyskole, 5000 Bergen (Norway)

(Received June 5, 1987; revision accepted October 1, 1987)

Storetvedt, K.M. and Otterå, L.E., 1988. Palaeozoic reconfigurations of north Scotland based on palaeomagnetic results from Orkney dykes. Phys. Earth Planet. Inter., 52: 243–255.

The camptonite and monchiquite dykes of the Orkney Islands, intruding Middle Devonian sedimentary strata, constitute two significantly different axes of magnetization. The inferred older of these remanences has an inclination corresponding to an Upper Devonian to early Carboniferous age while the inclination of the secondary magnetization is in line with the Kiaman overprint repeatedly recorded in Orcadian Basin rocks. However, the unique palaeomagnetic property of these dykes is that both magnetizations have mean declinations of ~ 170°, i.e., the palaeomagnetic axes are rotated ~ 20° anticlockwise with respect to the corresponding NW European reference fields. The data represent vital new information for the discussion of transcurrent motions along the Great Glen Fault. It is concluded that the Orkney dykes originated at a late stage of an Acadian age sinistral megashear (~ 600 km) at a time when the Northern Highlands block was positioned to the south of the Central Highlands. Partial remagnetization of the dykes took place at an early stage of Hercynian fault reactivation. The present declination discrepancy of both primary and overprinted magnetizations was brought about by a Hercynian dextral displacement on the Great Glen Fault (~ 300 km) leading essentially to the present juxtaposition of the Northern and Central Highlands.

1. Background and objectives

The Orkney Islands (Fig. 1a) are nearly exclusively built up of flagstones and sandstones belonging to the Middle and Upper divisions of the Old Red Sandstone (ORS) (Wilson et al., 1935; Mykura, 1976). This Devonian sedimentary succession, spanning some 5 km in stratigraphic thickness, rests on a regional granitic/metamorphic basement complex locally exposed around Stromness, West Mainland. The tectonic structure is characterized by numerous dissecting faults and very open folds (the main ones having a near N–S trend) providing mostly gently inclined strata (dips of the order of 10°).

Minor amounts of igneous rocks are scattered throughout the archipelago. The earliest phase of the igneous activity is represented by at least two lava flows intercalated in the upper section of the Middle ORS of East Mainland, Shapinsay and the Black Holm of Copinsay (Wilson et al., 1935). A second volcanic horizon occurs at the base of the Upper ORS of Hoy in association with a marked tectonic unconformity. From palaeomagnetic evidence it has been concluded that the Hoy volcanism and adjacent stratal disturbances were interrelated phenomena (Storetvedt and Meland, 1985). $^{40}Ar/^{39}Ar$ stepwise degassing analysis suggests that the Hoy volcanism dates from c. 370 Ma, while anomalously young conventional K–Ar

0031-9201/88/$03.50 © 1988 Elsevier Science Publishers B.V.

ages from the same rocks (varying between 333 and 300 Ma) are interpreted in terms of Ar loss during subsequent Permo-Carboniferous hydrothermal events (Halliday et al., 1977). The conclusions of the latter authors are fully supported by the palaeomagnetic evidence in that the ORS magnetization is found to be overprinted by a Permo-Carboniferous remanence component (Storetvedt and Meland, 1985).

A tectonic phase of upper Middle Devonian age is clearly established in Hoy, but in the John O'Groats Sandstone of NE Caithness (Fig. 1b) the stratal disturbances may locally post-date the well-established magnetization overprint (Storetvedt and Carmichael, 1979). It has been concluded therefore that the integral structural deformation of the Devonian strata of Orkney/Caithness includes both mid-Devonian and Permo-Mesozoic tectonic components (Storetvedt and Meland, 1985) probably related to phases of tectonic reactivation of the Great Glen Fault (Storetvedt, 1987). In fact, evaluation of declination discordances astride the Great Glen Fault suggests that it was the locus of major shear movements both in late Caledonian (Acadian) and in Hercynian times (see below).

The youngest igneous event in the Orkneys is represented by mostly thin (0.5–1 m), vertical lamprophyre dykes cutting the Middle ORS succession, but a few dykes do also penetrate into lower stratigraphic levels of the Hoy Sandstone (Upper ORS). It is this dyke swarm that is the object of the present palaeomagnetic study. K–Ar results from three of the dykes (camptonites) display an unsatisfactory scatter, the ages varying between 269 ± 11 and 245 ± 13 Ma (Flett Brown, 1975). By the same technique Halliday et al. (1977) arrived at a mean age of 235 Ma for a monchiquite dyke on Hoy. The importance of Permo-Mesozoic overprinting of Orcadian Basin rocks is now a

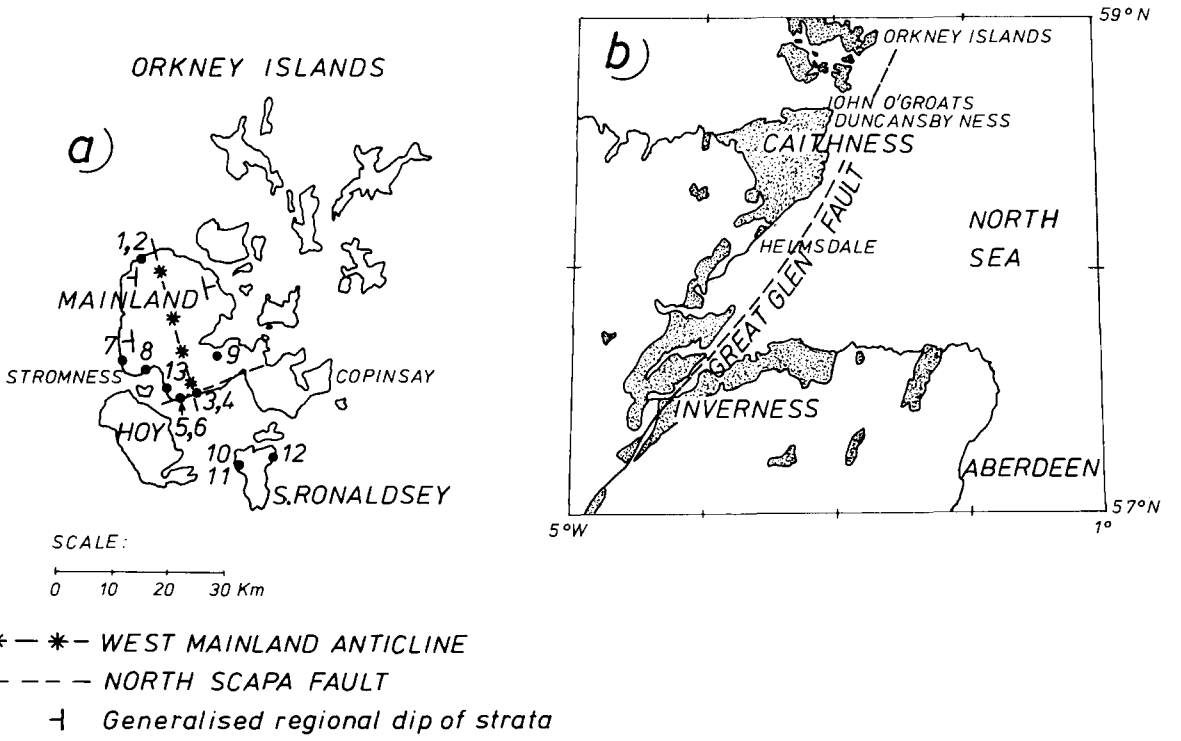


Fig. 1. Sketch map of the Orkney islands (a) showing geological distribution of the investigated dykes along with prominent tectonic features in the study area. Diagram (b) outlines part of the Orcadian Old Red Sandstone Basin (shaded) and shows the location of Orkney with respect to the Great Glen Fault and the areas of Caithness that are referred to in the text.

well-established fact (e.g., Tarling et al., 1976; Storetvedt et al., 1978; Storetvedt and Carmichael, 1979; Storetvedt and Meland, 1985). Therefore, the large age spread obtained from the Orkney dykes is most probably due to differential resetting of the radiometric system during post-cooling hydrothermal activity, probably associated with phases of post-Caledonian tectonic reactivation of the Great Glen Fault (Storetvedt, 1987). In the new palaeogeographic model of north Scotland (Storetvedt, 1987) the upper Middle Devonian tectonomagmatic event in Hoy is thought to mark the onset of a c. 600 km sinistral displacement on the Great Glen Fault. The question then arises whether the Orkney dykes were injected during this phase of megashearing. The palaeomagnetic results obtained strongly support this suggestion.

In a recent palaeomagnetic study of volcanics and sediments of the Eday Group of eastern Mainland Orkney, and the adjacent islands of Burray, South Ronaldsay, Black Holm of Copinsay, and Shapinsay, Robinson (1985) has reported directions of magnetization that in part must be considered anomalous and questionable in the context of Orcadian Basin palaeomagnetism. The results concerned are reversed but with a N–S smeared distribution, the average dip-corrected inclinations varying from +23.0 to +74.1, and examples given from thermal demagnetization show a pronounced tendency for the magnetization vectors to become either more steeply inclined (one sample even shows a swing into normal polarity) at high temperatures or to move towards a shallow reversed direction. The reported magnetization directions are therefore most probably unresolved dual-polarity remanences, and for that reason they are not included in the present palaeomagnetic discussion.

2. Sampling and laboratory procedures

The Orkney dyke suite can be seen in more than 200 exposures (Wilson et al., 1935), but, owing to an extensive Quaternary cover, the individual dykes can normally only be followed over short distances (5–50 m). However, the great majority of dyke exposures proved to be too weathered for palaeomagnetic sampling, and in addition many dykes were only exposed in fairly inaccessible coastal cliffs. These problems greatly reduced the number of actual sampling locations, and suitable material was only obtained from 13 dykes. The areal distribution of these sites is depicted in Fig. 1a along with main tectonic structures in the sampling region. The palaeomagnetic collection comprises 61 oriented hand-specimens. Magnetic orientation was used throughout, but for 26 of the samples a sun compass was also employed. The differences between sun and magnetic compass readings are random, and the average declination based on the two types of reading is

TABLE I

Rock collection details, Orkney dykes

Site	Samples	Location	Strike	Dyke width (m)	Rock type
1	OD 1– 5	Birsay Bay	070	~1	Camptonite
2	6–10	Birsay Bay	070	1.0	Camptonite
3	11–15	Ch. of Orphir	090	0.5	Camptonite
4	16–20	Ch. of Orphir	040	0.5	Camptonite
5	21–25	Midland Ness	170	~1	Monchiquite
6	26–29	Midland Ness	170	0.5	Monchiquite
7	30–34	Billia Croo	090	0.5	Camptonite
8	35–39	Garson, Strom.	054	~1	Camptonite
9	40–45	Rennibister	066	~2	Camptonite
10	46–48	Widewall Bay	060	0.8	Camptonite
11	49–52	Widewall Bay	060	1.0	Camptonite
12	53–57	Grim Ness	040	1.0	Monchiquite
13	58–61	Petertown	138	2.5	Monchiquite

10.4° W, i.e., practically identical to the regional declination value, with a standard deviation of 2.4°. Further collection details are given in Table I.

In the laboratory, cylindrical specimens, with a diameter and height of 1.9 cm, were cut for palaeomagnetic measurements. Demagnetization was either by alternating field (AF) using a two-axis tumbler, or by the thermal technique employing a Schonstedt TM-1 as well as a non-commercial set-up. After each demagnetization step the remanence parameters of each specimen were determined using a Digico or a Molspin magnetometer. For analytical purposes, a combination of stereoplots, orthogonal vector diagrams and vector subtraction has been used.

3. Laboratory data

A total of 144 specimens have been subjected to AF or thermal demagnetization. All of the palaeomagnetic results here constitute linear segments in orthogonal vector diagrams, and most of them represent end-point results. Around 50% of the investigated specimens exhibit more or less systematic vectorial swings with increasing demagnetization, ending in an erratic stage that precludes the end-point magnetization from being determined. Some of these swinging specimens move along well-defined great circle segments, suggesting that the remanence is that of a two-component system, but only occasionally is there a 'window' in the demagnetization spectrum for which only one remanence component is being erased. For a minor series of samples a combination of AF demagnetization followed by thermal treatment was attempted. On the whole, this procedure was not very successful in that an erratic (unstable) stage appeared at an early stage of thermal demagnetization.

A broad range of demagnetization data, outlining the magnetic structure of individual specimens or dykes, is depicted in Figs. 2–4. On the whole, the Orkney dykes provide a typical multicomponent case. In addition to a soft component, frequently aligned along the present Earth's field (PEF), there are two characteristic directional populations of reverse polarity, magnetizations A ($D_m = 172$, $I_m = -24$) and B ($D_m = 169$, $I_m = +4$), that are clearly discriminated by direction.

In a nutshell, the results of sites 1 and 2 (Fig. 2) give a proper presentation of the characteristic directions of the dyke system. These sites represent two parallel and closely located dykes at Birsay Bay, probably constituting two locally separated branches of the same intrusive body. In the majority of specimens from these two sites the A-component predominates, but it is frequently found that upon demagnetization of the last 15% or so of the magnetic moment the remanence vector moves into the B-direction. For specimen 9-F2 (Fig. 2i) demagnetization in the 24–37 mT range describes a great circle arc within which the various difference vector directions cluster (i.e., defining straight lines in orthogonal vector plots). The erased magnetization in this range, the direction of which is shown by the triangle in Fig. 2i, is the A-component, and the end-point magnetization (the 37–49 mT range) corresponds to the B-remanence. It is an important observation that in cases where the A- and B-magnetizations coexist in the same specimen, the B-component is always the one of higher stability, e.g., 1-C2 (Fig. 2b, j), 6-C2 (Fig. 2d), 9-B3 (Fig. 2h, n) and 9-F2 (Fig. 2i). A similar behaviour is seen in samples 7-B2 and 8-B3 (Fig. 2e, f), but in these cases the magnetically unstable stage occurring above 60 mT (not included in the diagram) prevented the B-magnetization from being experimentally confirmed. Samples 4-B4 (Fig. 2a, k) and 7-B 1 (Fig. 2g, l) appear to represent cases with only one of the characteristic magnetizations present. Figure 2c shows a stereographic projection of the directional trends of four specimens for which univectorial magnetizations could not be determined. Note the progressive northerly movement of the remanence vector for specimens 1-B3 and 9-E2, suggesting the presence of a high coercive force/high blocking temperature magnetization of normal polarity. In the present study there is a relatively large number of such cases, but it has not been possible to establish the direction of this normal component(s). All A and B directions extracted from sites 1 and 2 are depicted in Fig. 2o.

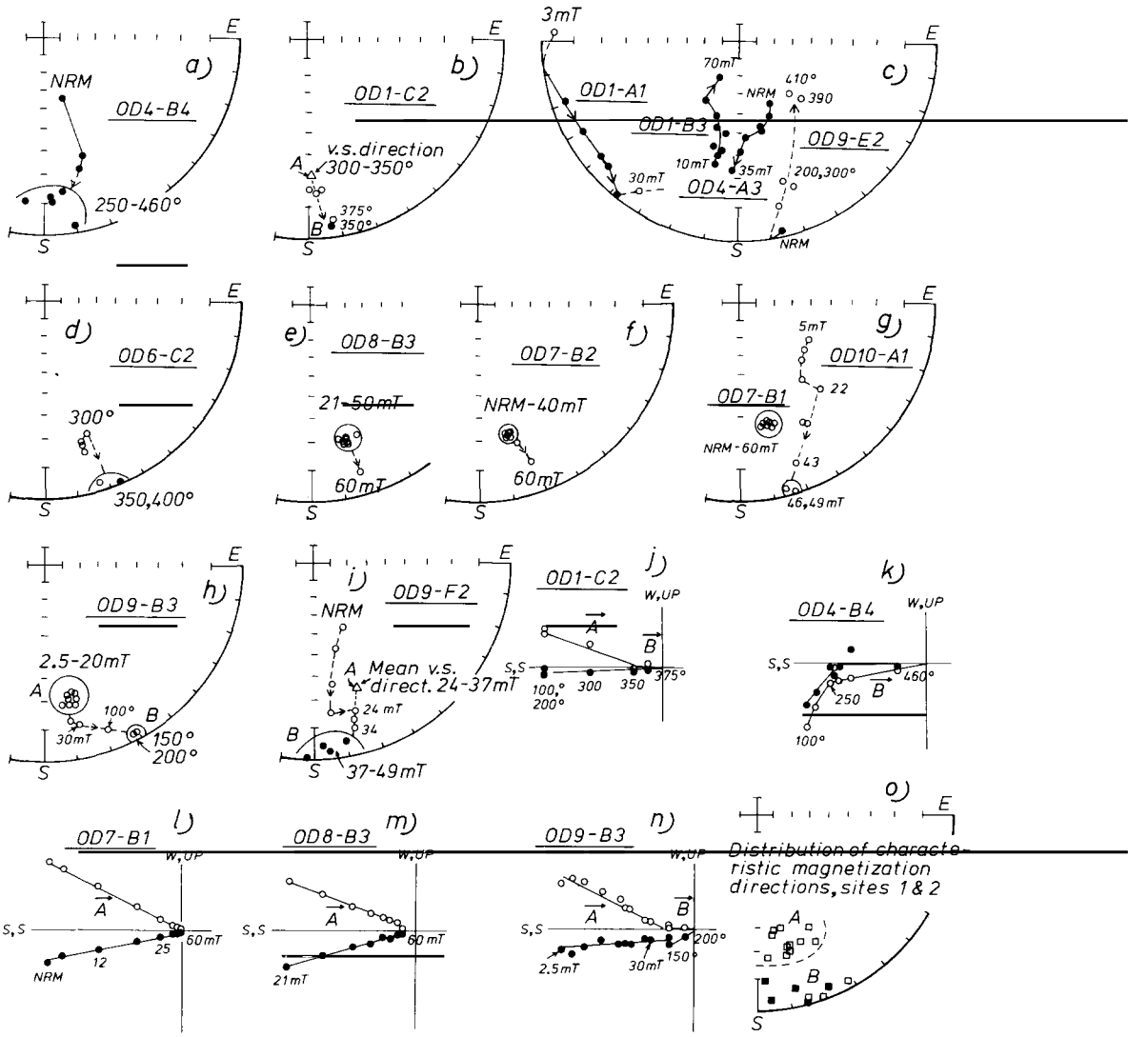


Fig. 2. Examples of demagnetization results from sites 1 and 2. Note the co-existing A- and B-magnetizations in various samples (cf., OD1-C2, 9-B3 and 9-F2). Plots are stereographic projections (Figs. a–i, o), on which ● (○) represent directions in the lower (upper) hemisphere, or in orthogonal vector diagrams (Figs. j–n) for which projections in the vertical and horizontal planes are marked by open and closed symbols respectively. Alternating peak field values are in mT.

Other sampling locations provide essentially one-component cases. Thus, site 5 (Fig. 3a–d) is dominated by the A-magnetization, and sites 7 and 9 are governed by the B-component (Fig. 3e–l). Material from site 12 has given five fairly well-defined specimen results, four A-directions and one B-direction. The predominance of the A-magnetization of this site is further demonstrated by the convergence of remagnetization circles for four specimens (54-D2, 55-D, 56-D and 57-E3) for which univectorial remanences have not been experimentally/analytically determined. For space-saving purposes, most vector plots of Figs. 2 and 3 display only the trajectories of the two characteristic magnetizations, i.e., excluding that fraction of the NRM with the lowest stability. To compensate for this possible loss in complete magnetization analysis, Fig. 4 demonstrates full

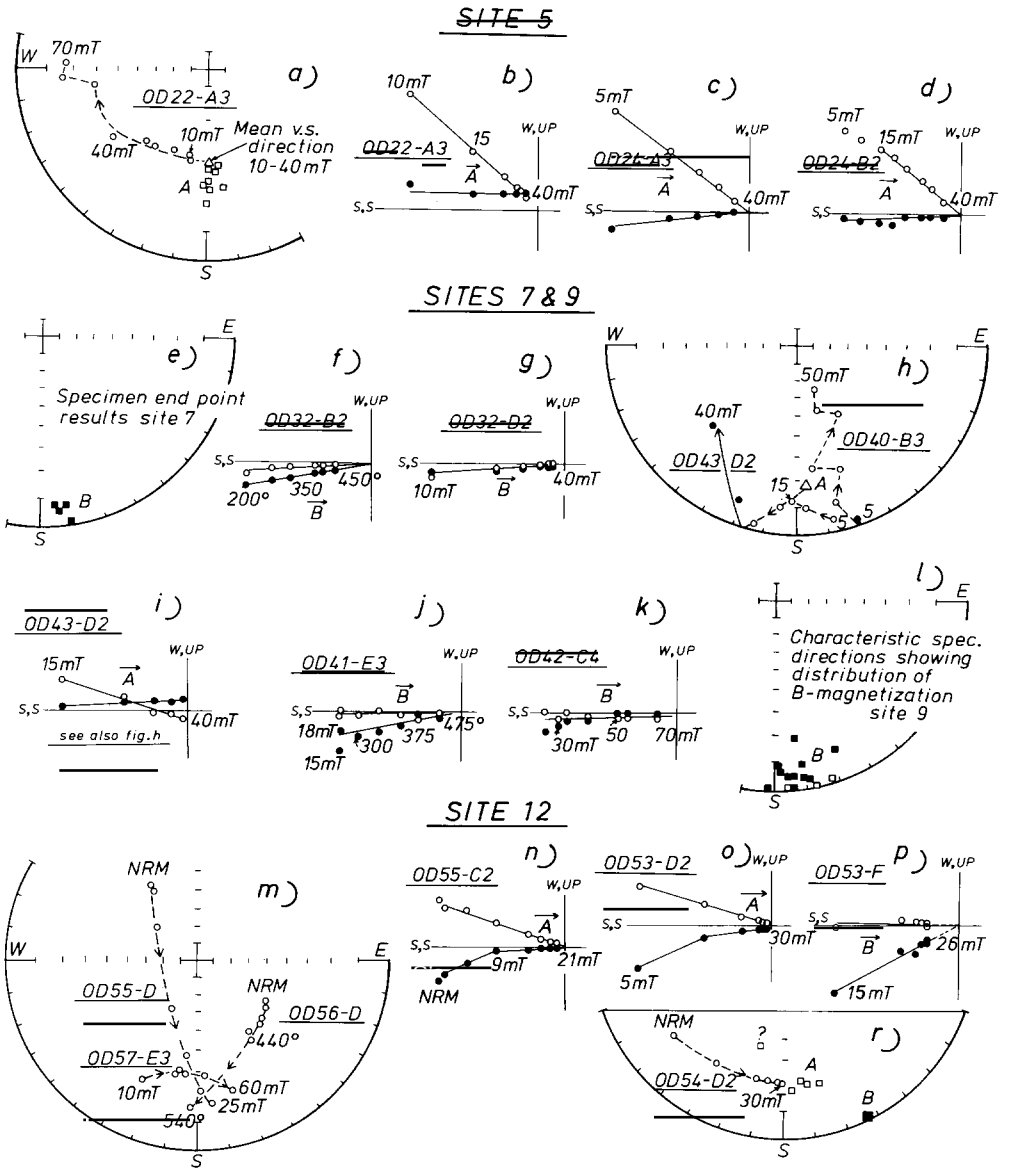


Fig. 3. Further demagnetization results from Orkney dykes. △ represent average erased vectors along great circle demagnetization paths for range indicated (cf., OD22-A3, a; OD43-D2, h). For these specimens the magnetization entered the erratic stage before the most stable component (normal polarity) had been determined. In such cases the vector trajectories of orthogonal plots will of course bypass the origin (i). Characteristic remanence directions for individual sites (a, e, l, r) are shown by square symbols. (m, r). Some demagnetization trends for which the last step before the erratic stage brings the remanence vector into the A-component area. Specimens of this kind have not given results that enable the extraction of individual palaeomagnetic components, but they add further confidence to the characteristic results obtained from other specimens. Conventions as for Fig. 2.

demagnetization diagrams for both A- and B-component specimens. In all, the present study has provided a total of 42 A-directions and 32 B-directions.

The inferred multiphase magnetization, a feature very different from that which would be expected if the dykes had essentially retained their original cooling remanence, should somehow be

reflected in the texture and composition of the Fe–Ti oxides. Reflection microscopy shows an apparent absence of ilmenite lamellae in the titanomagnetite (TM) grains, suggesting that high-temperature oxidation is at most only developed to a very modest extent. After cooling, the homogeneous TM particles are assumed to have had low Curie points (T_c) related to a relatively high Ti content. In all probability, the original TM grains were therefore fairly vulnerable to low-temperature alteration, and the magnetomineralogical evidence concurs with this assumption. Thus, the saturation magnetization (J_s) versus temperature shows mostly well-defined curie points in the

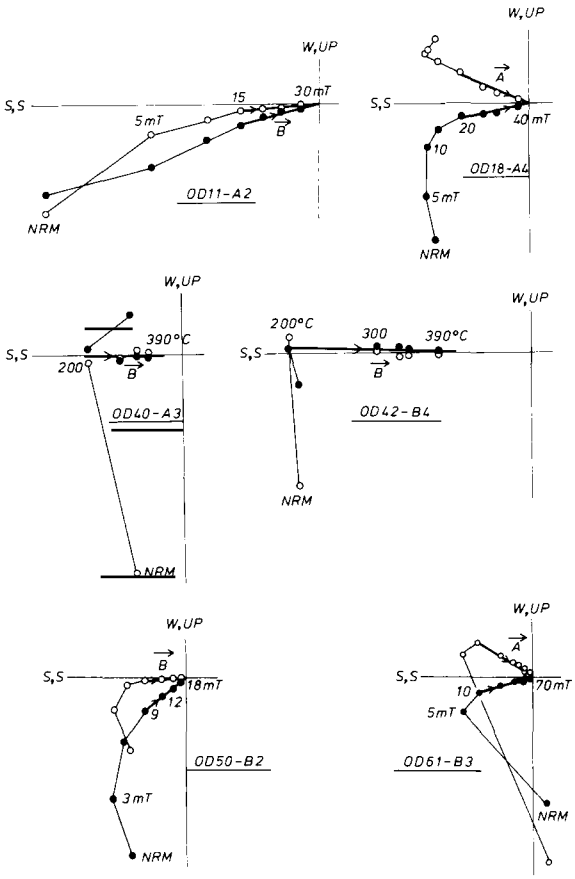


Fig. 4. Orthogonal vector diagrams showing magnetization changes versus progressive demagnetization for some A- and B-component specimens not included in Figs. 2 and 3. AF-treated specimens provide the most complete demagnetization patterns as thermal treatment above c. 400 °C frequently gave erratic results. Conventions as for Fig. 2.

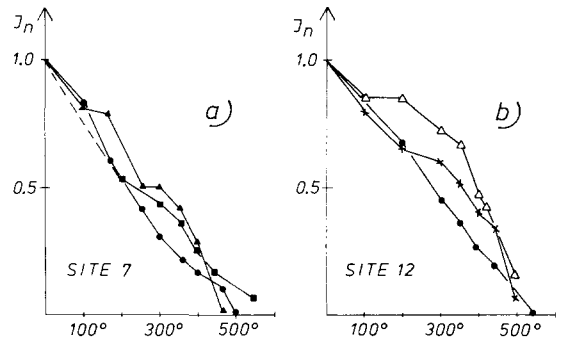


Fig. 5. Examples of thermal decay patterns produced by plotting the magnetic moment measured at room temperature, after successive steps of heating and cooling in field-free space. Site 7 (a) is characterized by the B-magnetization while site 12 (b) is dominated by the A-component.

450–580 °C range, and, in addition, ill-defined 'Curie temperatures' in the 200–400 °C range, associated with pronounced J_s–T irreversibility, is a predominant feature notably for material from the contact zones. Microscopically the TM grains exhibit partial replacement by haematite and maghaemite as well as displaying granulation textures (Ade-Hall et al., 1971). It appears most likely therefore that low-temperature alteration through oxidation/conversion processes has produced phases such as maghaemite, magnetite and Ti-poor TM or titanomaghaemite which apparently constitute the present remanence-carrying mineralogy. It is also an important observation that the thermal decay patterns of the natural remanent magnetization (see Fig. 5) delineate blocking temperature spectra that are very similar for both A- and B-component material.

Figure 6 depicts the in-situ distribution of all individual directions of the two characteristic magnetizations. The two populations do not overlap and the two mean directions are significantly different at the 95% significance level (cf., Table II). This discrimination between the mean directions is also very clear from Watson's Test (Watson, 1956). Some preliminary A-magnetization data, based on an early phase of demagnetization, have been reported elsewhere (Storetvedt, 1987). However, in the final analysis of all the results (presented here) three of the previously reported

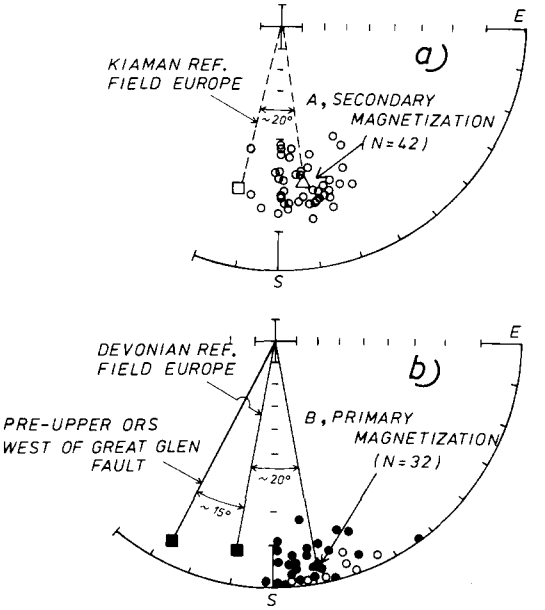


Fig. 6. Stereoplots showing the distribution of the two fossil magnetization populations, components A (a) and B (b), in conjunction with relevant palaeomagnetic reference axes. The overall palaeomagnetic directions for the dykes are shown by △. Note that both remanence groupings are rotated anticlockwise with respect to the reference fields. These differences are accounted for by the tectonic model of Fig. 7. Diagram conventions as for Fig. 2.

specimen directions were found to be of questionable reliability and are therefore not included in Fig. 6.

4. Magnetization history in a tectonic framework

The component analysis outlined above suggests that the Orkney dykes have recorded two major magnetization 'events'. The overall in-situ directional results of the two magnetizations (A and B) are summarized in Table II in conjunction with other relevant data from ORS rocks of the Orcadian Basin. We note interesting similarities such as (1) the dual magnetization structure and (2) the fairly constant inclination of the various B-magnetizations. When the two remanence components co-exist in individual specimens it is always the B-component that has the higher stability (Storetvedt et al., 1978; Storetvedt and Meland, 1985; present study). This important observation, along with conventional palaeopole considerations, has been taken to favour a secondary origin of the A-magnetizations. However, the A-remanences show some differences in inclination, suggesting that their times of origin vary (Storetvedt and Meland, 1985). Thus the John O'Groats Sandstone and the Duncansby Volcanic Neck, NE Caithness, appear to have acquired their A-magnetizations in Mesozoic time (Storetvedt et al., 1978; Storetvedt and Carmichael, 1979) while inclination and polarity evidence indicate that rocks such as the Helmsdale granite (Torsvik et al., 1983), the Hoy volcanics (Storetvedt and Meland, 1985), volcanics and sediments of the Eday Group, Southern Orkney (Robinson, 1985)

TABLE II

Overall palaeomagnetic results from the Orkney dykes in comparison with other relevant data of the Orkney/Caithness region. Statistics according to Fisher (1953)

Formation	Pal. group	D	I	α_{95}	K	Pole
'Original' magnetizations						
Orkney dykes (present study)	B	169	+4	3.1	67	189.5° E, 28.4° N
Hoy lavas (Storetvedt and Meland, 1985)	B	208	+9	5.5	43	146.3° E, 22.7° N
John O'Groats Sandstone (Storetvedt and Carmichael, 1979)	B	209	+6	10.4	18	144.5° E, 24.1° N
Lower Caithness ORS (Storetvedt and Torsvik, 1983)	B	205	+3	6.4	20	148.5° E, 27°·0N
Duncansby Neck (Storetvedt et al., 1978)	B	205	+8	6.3	28	149.1° E, 24.3° N
Overprinted Kiaman age magnetizations						
Orkney dykes (present study)	A	172	−24	3.3	51	188.9° E, 43.2° N
Hoy lavas (Storetvedt and Meland, 1985)	A	176	−31	4.2	206	182.2° E, 47.7° N
Helmsdale granite (Torsvik et al., 1983)	A	183	−31	6.0	29	171.8° E, 48.9° N
Volcanic and sedimentary rocks SE Orkney (Robinson, 1985)	A	191	−20	5.9	17	163.2° E, 40.6° N

and the Orkney dykes (present study) have A-components that were imposed in the Permo-Carboniferous (cf., Storetvedt (1987) for general discussion). The inclination of the B-magnetizations corresponds to the palaeomagnetic field that characterizes the Devonian–Lower Carboniferous (Storetvedt, 1967, 1987). However, the Orkney dykes are unique in one respect; both palaeomagnetic axes show somewhat anomalous declinations. Figure 6 poses the actual situation. Both the A- and B-magnetizations are rotated c. 20° anticlockwise relative to the corresponding NW European reference fields (corresponding to poles at 163° E, 44° N, and 160° E, 20° N, respectively) (cf., for example, Storetvedt, 1967) and there is no correlation between characteristic magnetizations and strike directions of individual dykes.

We have so far assumed that it is the in-situ remanence directions that are of palaeomagnetic relevance, but a simple fold test may add further substance to this assumption. The area of West Mainland, in which ten of the 13 investigated dykes are located, is tectonically controlled by the West Mainland Anticline. Figure 1a shows the axial trace of this fold structure which provides gently westward-dipping strata on the western flank and eastward-dipping strata of similar magnitude (c. 10°) on the eastern flank. Fortunately, the two dykes with the most clear-cut B-magnetization, numbers 7 and 9, are located on opposite flanks of the West Mainland Anticline. In the in-situ state the two mean directions (cf. Fig. 3e, l, for within-site distribution) are identical; D, I, $\alpha_{95} = 172, +4, 3.1°$ for dyke 7 and 172, +4, 4.2° for dyke 9. Owing to the shallow B-magnetization, combined with a regional strike direction that is very close to the remanence declination, the directional response to tectonic correction will be small. However, adjustment of the in-situ B-magnetization of dykes 7 and 9 by 10° back-tilting (of the sedimentary strata) increases the separation of the two mean directions from 0 to c. 6°. Therefore, this result tentatively supports the geological evidence that the Orkney dykes post-date the local structural deformation (Wilson et al., 1935; Mykura, 1976).

In addition to the palaeomagnetic declination discrepancies for the Orkney dykes it has long been recognized that (using the same reference frame) suggested pre-Upper ORS magnetizations west of the Great Glen Fault exhibit a c. 15° westward declination discordance (Storetvedt, 1974; Storetvedt et al., 1978; Storetvedt and Carmichael, 1979; Storetvedt and Meland, 1985, etc.). In explaining these features one has to realize that both magnetization axes of the Orkney dykes are anomalous in the European palaeomagnetic context, so any hypothetical explanation in terms of erratic field behaviour must therefore be discounted. On the other hand, the palaeomagnetic features of Orcadian Basin rocks can be readily accommodated in a tectonic model involving large-scale strike-slip displacements on the Great Glen Fault, implying relative rotations of the Northern Highlands of Scotland. The model has been presented in detail elsewhere (Storetvedt, 1987), and reference is made to that paper for geological and geophysical details. We shall here limit our discussion to the question of how the experimental results on the Orkney dykes can be fitted into this regional palaeomagnetic/tectonic framework. Figure 7 depicts the inferred palaeogeographic development. The Northern Highlands were originally positioned a few hundred kilometres north of their present location but in the late Middle Devonian a c. 600 km sinistral movement on the Great Glen Fault brought the Northern Highlands block to the south of the Central Highlands (Fig. 7a). The sinistral displacement led to a c. 35° clockwise rotation of pretectonic magnetizations of the Northern Highlands relative to corresponding NW European data (Storetvedt, 1987). At the final stages of the Acadian age megashear, local conditions gave rise to the formation of the camptonite/monchiquite dyke swarm in Orkney. These narrow dyke intrusions, which probably cooled very quickly, are likely to have had an original Fe–Ti oxide mineralogy with a high Ti content. The chemical instability of such mineral phases in a regional late–post-magmatic thermochemical environment (brought about by the tectonomagmatic processes) is likely to have provided a certain prolongation of the acquisition time of the 'original' magnetization (the B-component). Therefore, by the time the B-magnetization of the Orkney dykes had been frozen in, the

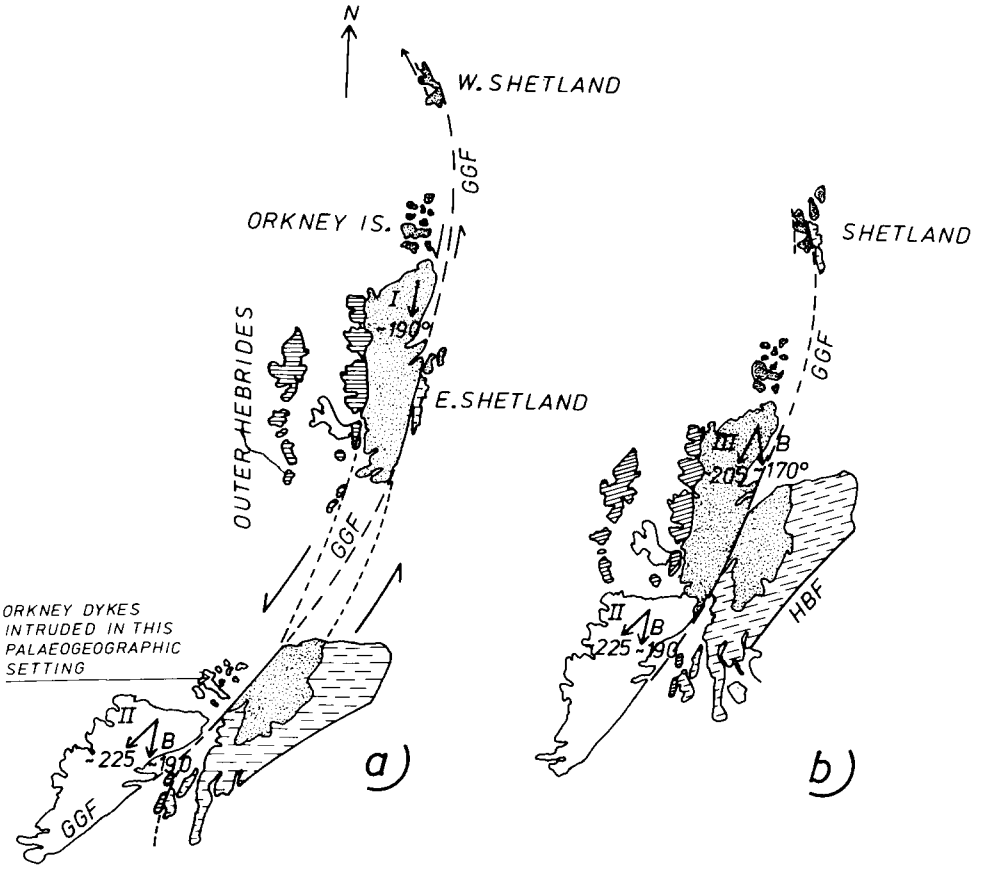


Fig. 7. Palaeozoic reconfigurations of north Scotland accommodating the observed palaeomagnetic discordances. The Orkney dykes were injected at a late stage of a c. 600 km sinistral shear, corresponding to ~35° relative clockwise rotation of the Northern Highlands, along the Great Glen Fault in Acadian (Upper Devonian) time (a) in a 'regional' hydrothermal regime that provided a certain amount of late-magmatic low-temperature alteration during acquisition of the primary magnetization (component B). In addition to the original remanence declination of the Orkney dykes (marked by arrow B) (a) also delineates the inferred pre-tectonic ORS magnetization relative to the Northern Highlands before and after the Acadian megashear, i.e., stages I and II with declination ~190° and ~225° respectively. The Great Glen Fault was reactivated in Hercynian time and at an early stage of this faulting a new phase of associated hydrothermal activity impressed a second event of low-temperature alteration and partial remagnetization into the dykes, affecting particularly magnetic grains that had 'escaped' the earlier Acadian age alteration processes. At an 'embryonic' stage of the Hercynian movement the A- and B-magnetizations are considered to have had overall declinations close to that of their corresponding European reference fields, i.e declination ~190°, but the subsequent dextral shear (b) rotated both magnetizations c. 20° anticlockwise to their present more easterly positions (cf., also Fig. 6). On the other hand, the two-phase rotation (sinistral followed by dextral) did not bring the Northern Highlands back to their original relative pre-Acadian geographical situation so that ORS magnetizations of pre-Acadian age from the Northern Highlands now display a westward discordance in declination of c. 15° (cf., Fig. 6b).

average remanence declination is thought to have been close to that of the NW European ORS reference field (i.e., $D_m \approx 190$), while the pre-tectonic ORS magnetizations of the Northern Highlands had by then a mean declination of c. 225° (owing to clockwise, pre-injection tectonic rotation of c. 35°). The indicated high coercive force/high blocking temperature remanence of normal polarity (and probably of low intensity), and for which the direction of magnetization has

not been established (cf., section 3), is most probably a B-component as the Devonian–Lower Carboniferous field had a dual-polarity structure.

It seems that the Northern Highlands block remained in the southern position until Hercynian times, during which the Great Glen Fault underwent a major dextral transcurrent movement. It is assumed that the Hercynian fault rejuvenation released a new phase of thermochemical activity and associated secondary changes of Fe–Ti oxides, giving rise to the Kiaman age A-component as well as to extensive radiometric age resetting (the K–Ar age spread can be explained by variations in Ar retentivity and by local differences in the intensity of overprinting processes). This Hercynian alteration was probably particularly effective in grains that had escaped the Acadian age secondary processes. This probably explains why the A- and B-magnetizations on the whole tend to have similar blocking temperature ranges. On the other hand, for specimens with coexisting A- and B-remanences, the latter component is always associated with the higher blocking temperatures. This may be taken as evidence for a lower temperature rise during the Hercynian tectonic phase than during the Acadian phase. This explanation appears reasonable, taking into account that Acadian displacement figure (~ 600 km) was clearly in excess of the Hercynian figure (c. 300 km) (Storetvedt, 1987).

The partial remagnetization during the Hercynian tectonic phase could of course have been impressed at any stage of the faulting process, the timing and relative intensity of the secondary magnetization depending on factors such as proximity to the Great Glen Fault and other major fractures (from which the thermochemical activity spreads out), porosity and degree of fracturing of the rocks, state of oxidation of the oxide mineralogies, etc. The Orkney dykes seem to have acquired the bulk of their Kiaman overprint at an early stage of the Hercynian shearing event. The Eday sediments of Southern Orkney (Robinson, 1985) seem to provide a case of late-stage magnetic resetting (i.e., having their A-directions close to that of the NE European Kiaman field), while sediments from the same group in Hoy (Storetvedt and Meland, 1985) appear to have acquired their secondary magnetization at an early stage of the faulting (Storetvedt, 1987). Thus, secondary magnetization in different rock bodies, linked to faulting and associated plate rotation, will naturally describe a smear of their mean declination (owing to differences in time of remanence acquisition), the maximum deviation from the corresponding reference field approximating to the tectonic rotation in question. It is inferred that the dextral Hercynian displacement along the Great Glen Fault (Fig. 7b) led to a relative rotation of the Northern Highlands of the order of 20°. This anticlockwise rotation did of course also affect the pre-tectonic ORS (pre-upper Middle Devonian) magnetization, changing its declination from c. 225° to c. 205°. Thus, the earlier declination discordance of c. 35° in ORS data astride the Great Glen Fault was reduced to c. 15°. Therefore, when comparing pre-late Middle Devonian palaeomagnetic data across the fault, only a modest discordance of 15° will now be seen. However, the hidden information about the true rotation/displacement in Acadian time can, for example, be unravelled by studying syntectonic ORS magnetizations within rocks adjacent to the Great Glen Fault (see, for example, Storetvedt (1987)), but the easterly direction of magnetization for the B-component of the Orkney dykes does also provide important independent information in favour of a significant mid-Palaeozoic (pre-mid Carboniferous) movement along the fault.

Though the prominent faults cutting the ORS of the Orkneys were already in existence during the ORS deposition (Mykura, 1976), these structures also seem to have played an important role in the Hercynian remagnetization of the dykes. One can imagine that during tectonically active periods the thermochemical environment along these lines of weakness was more active than elsewhere in the region. For example, sites located close to the North Scapa Fault (see Fig. 1a) exhibit a predominance of the Hercynian age A-magnetization (cf., site 5, Fig. 3) while sites more distant from prominent faults (for example, numbers 7 and 9) are basically characterized by the original (Upper Devonian) B-magnetization.

5. Conclusions

From the present palaeomagnetic study of the Orkney dykes the following conclusions can be drawn:

(1) The rocks possess two well-defined principal remanence components, A ($D = 172$, $I = -24$, $\alpha_{95} = 3$) and B ($D = 169$, $I = +4$, $\alpha_{95} = 3$). This two-axis magnetization is in line with the characteristic palaeomagnetic build-up repeatedly reported from Orcadian Basin rocks, but a new and important observation is that both remanences exhibit a c. 20° anticlockwise rotation (in declination) with respect to relevant NW European reference fields. However, the declination discordance between the B-magnetization of the Orkney dykes and that of local pre-Upper Devonian rocks (west of the Great Glen Fault) is even more pronounced, amounting to at least 35°. The only realistic interpretation of these palaeomagnetic discrepancies appears to be in terms of tectonic rotation, and a two-stage relative rotation of the Northern Highlands is invoked to explain the palaeomagnetic inconsistencies.

(2) We conclude that the B-magnetization was imposed in late Middle–Upper Devonian (i.e., in Acadian time), during which the Great Glen Fault was subjected to a sinistral movement of the order of 600 km. This displacement was associated with a c. 35° clockwise rotation of the Northern Highlands. The Orkney dykes are thought to have been injected during a late stage of this megatectonic event. At the time of dyke intrusion the Northern Highlands block was positioned to the southwest of the Central Highlands.

(3) In Kiaman time (Middle Carboniferous–Upper Permian) the Hercynian diastrophism in western Europe reactivated the Great Glen Fault, and a new phase of thermochemical activity spread out from the fault zone, leading to partial remagnetization (the A-component) of adjacent rock formations. The Kiaman overprinting seems to have been impressed at an early stage of the Hercynian faulting, which was of dextral nature. The dextral shear (of the order of 300 km) led to an anticlockwise rotation of the Northern Highlands of around 20°, terminating in essentially the present geographic situation. This explains why the characteristic magnetization axes of the dykes are misaligned in the anticlockwise sense with respect to the relevant NW European palaeofields. The Hercynian thermochemical phase along the Great Glen Fault led to extensive but somewhat variable radiometric age resetting; the true Acadian age (say c. 350–370 Ma) of the dykes was lost, and mixed K–Ar dates in the Permian age range became radiometric end-products.

Acknowledgements

The field sampling for this study was financed by K. Birkelands Fond. We are most grateful to Mr. K. Breyholtz for competent laboratory work and for skilful preparation of the diagrams, to Mrs. E.J. Eriksen for typing of the manuscript, and to Drs. N. Bonhommet, E.R. Deutsch and M. Westphal for constructive comments.

References

Ade-Hall, J.M., Palmer, H.G. and Hubbard, T.P., 1971. The magnetic and opaque petrological response of basalts to regional hydrothermal alteration. Geophys. J. R. Astron. Soc., 24: 137–174.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London Ser. A, 217: 295–305.

Flett Brown, J., 1975. Potassium–argon evidence of a Permian age for the camptonite dykes, Orkney. Scott. J. Geol., 11: 259–262.

Halliday, A.N., McAlpine, A. and Mitchell, J.G., 1977. The age of the Hoy lavas, Orkney. Scott. J. Geol., 13: 43–52.

Mykura, W., 1976. British Regional Geology: Orkney and Shetland. HMSO, Edinburgh, 149 pp.

Robinson, M.A., 1985. Palaeomagnetism of volcanics and sediments of the Eday Group, Southern Orkney. Scott. J. Geol., 21: 285–300.

Storetvedt, K.M., 1967. A synthesis of the Palaeozoic palaeomagnetic data for Europe. Earth Planet. Sci. Lett., 3: 444–448.

Storetvedt, K.M., 1974. A possible large-scale sinistral displacement along the Great Glen Fault in Scotland. Geol. Mag., 111: 23–30.

Storetvedt, K.M., 1987. Major late Caledonian and Hercynian shear movements on the Great Glen Fault. Tectonophysics, 143: 253–267.

Storetvedt, K.M. and Carmichael, C.M., 1979. Resolution of superimposed magnetization in the Devonian John O'Groats Sandstone, N. Scotland. Geophys. J. R. Astron. Soc., 58: 769–784.

Storetvedt, K.M. and Meland, A.H., 1985. Geological interpretation of palaeomagnetic results from Devonian rocks of Hoy, Orkney. Scott. J. Geol., 21: 337–352.

Storetvedt, K.M. and Torsvik, T.H., 1983. Palaeomagnetic re-examination of the basal Caithness Old Red Sandstone; aspects of local and regional tectonics. Tectonophysics, 98: 151–164.

Storetvedt, K.M., Carmichael, C.M., Hayatsu, A. and Palmer, H.C., 1978. Palaeomagnetism and K/Ar results from the Duncansby volcanic neck, northeastern Scotland: superimposed magnetizations, age of igneous activity and tectonic implications. Phys. Earth Planet. Inter., 16: 379–392.

Tarling, D.H., Donovan, R.N., Abou-Deeb, J. and El-Batrouk, S.I., 1976. Palaeomagnetic dating of haematite genesis in Orkney basin sediments. Scott. J. Geol., 12: 125–134.

Torsvik, T.H., Løvlie, R. and Storetvedt, K.M., 1983. Multicomponent magnetization in the Helmsdale granite, North Scotland; geotectonic implication. Tectonophysics, 98: 111–129.

Watson, G.S., 1956. Analysis of dispersion on a sphere. Mon. Not. R. Astron. Sco., Geophys. Suppl., 7: 153–159.

Wilson, G.V., Edwards, W., Knox, J., Jones, J.S.B. and Stephens, J.V., 1935. The geology of the Orkneys. Mem. Geol. Surv., 205 pp.